Introduction

Today, we will discuss what gave rise to the adoption of automation and what they achieved with automation. We will also learn how hyper-automation came into existence and how automation evolved with the industry's endeavor to achieve productivity and effectiveness.

Scaling beyond RPA

RPA, aka Robotic Process Automation, is an uninvolved integration technology designed to automate normal, repetitive, and predictable processes through coordinated UI interactions that replicate human behaviors.

> *According to Gartner, complementary technologies will provide value to 80 percent of RPA-centric automation initiatives by 2022.*
>
> *While 33% of respondents at a recent CIO webinar said they were just getting started with RPA, over 30% said they had already begun using modern technologies such as process mining, machine learning, and ingestion engines to boost automation success.*

The first quarter of 2022 is over and it is evident that RPA-centric automation is adopted across most industries to automate siloed departments of the organization.

There are two types of RPA-assisted and unassisted.

> **Assisted:** Goals achieved with a combination of bots and humans. Bots power separate desktops, and the human worker performs some parts of the activity while the bot handles the more time-consuming or technically tricky aspects of the process.
>
> **Unassisted:** Bots are installed on a centralized server and sometimes manually controlled. This sort of RPA technology may automate end-to-end operations and process scheduling from a single control point.

However, the organizations are not satisfied with siloed automation. Instead, they want everything integrated and streamlined to have a holistic view.

So, what's the next level of automation that organizations are chasing? It is hyper-automation. Let us learn the nuances of hyper-automation.

1. What is Hyperautomation?

Gartner, technological research and consulting firm, acknowledged 'Hyper-automation' as the top 10 technologies to look for in 2022. According to some tech pandits, RPA and IA are the subsets of hyper-automation.

Hyperautomation refers to the "combination of automation technologies with several machine learning applications and packaged software" leveraged to complete various levels of a task.

RPA vs. Intelligent Automation vs. Hyperautomation

	RPA	**IA**	**Hyper-automation**
Type of Task	Automation of repetitive & mundane tasks	Automation of tasks that involve logic	Automation of complex tasks
Human Involvement	Requires more	Requires less	Less to negligible
Understanding of Data	Structured Data	Logical data	Unstructured & complex data
Type of Solution	Simple task-based solutions	Logical task-based solutions	Complex process simplification
Process of Working	One type of task	One type of sector	Integrates everything & provides holistic view
Implementation Difficulty	Low	Moderate	High

Technology Components of Hyperautomation:

As mentioned above, we will discuss the evolution of automation. The development of automation started with one technology and kept stacking one after another to make it work like an end-to-end and agile platform that incorporates new information and workflows iteratively and incrementally.

The technologies incorporated to achieve hyper-automation are:

Artificial Intelligence(AI): Machine learning in artificial intelligence replicates human intellect. First, the computer learns how to respond to certain activities & creates a propensity model using algorithms and historical data. After that, propensity models will begin to make predictions.

iBPMS: An integrated business process management system (iBPMS) is a collection of technologies working together to coordinate humans and machines in delivering business processes. According to the business rules, an iBPMS allows organizations to model, implement, and execute sets of interconnected processes.

Even though it has become less popular with the introduction of RPA, it is still a helpful tool for process automation.

Ingestion Engines: Ingestion engines bring in data for process mining, decision modeling, machine learning, and business process management.

Process Mining: Process mining helps analyze a company's processes to acquire a comprehensive understanding. Process mining offers many applications, from conventional ones like 'process optimization' to industry-specific ones like 'risk detection in audits.'
Process mining is necessary for process simplification and comprehension, which are critical facilitators of hyper-automation.

Decision modeling: In hyper-automation, decision modeling models the logic of a decision formulated in multiple processes.

RPA: RPA is said to be at the core of hyper-automation. It helps automate mundane and repetitive business processes.

iPaaS: iPaaS helps connect software applications deployed in different environments.

Analytics: Analytics component plays a vital role as it helps measure and demonstrate the ROI of automation and its impact.

UX: UX is an essential component in providing a seamless engagement between automation and the user experience.

Orchestrated use of multiple technologies. These include artificial intelligence (AI), machine learning (ML), chatbots/conversational platforms, and robotics process automation (RPA)

2. What is the hype behind hyper-automation?

The pandemic brought a revolution of digital transformation and almost all enterprises prioritized their goals for digital adoption. Hyperautomation is one of those tech trends that has gotten everyone's attention.

Hyperautomation is rapidly making rounds across industries regardless of the size of the enterprise. According to Business World industry estimates, the worldwide hyper-automation market will be worth $600 billion by the end of 2022.

- **One of the top digital trends for 2022**

Gartner highlights technological trends that are crucial to businesses every year. This year's list includes 12 strategic themes that will help CEOs achieve growth, digitization, and efficiency while also positioning CIOs and IT executives as strategic partners in the firm.
Hyper-automation made it to the list.

- **The inclination towards hyper-automation**

According to David Groombridge, VP Analyst at Gartner, "CEOs see the need to drive digital business adoption and are looking for more direct digital ways to communicate with their consumers."

Why?

They want to be efficient and maintain profits and cash flow while keeping an eye on potential economic concerns, and this is what hyper-automation is enabling for the next digital age.

3. Is your company ready for significant change? (Checklist)

If your company wants to expand automation, you should first determine your objectives and how greater automation will help achieve the goals.

Let's look at some variables that might help you determine whether a particular industry or business is a good fit for hyper-automation. If these crucial aspects are left unconsidered, the advantages of hyper-automation cannot be optimally achieved.

- Robust business process:

Every use case should have a solid business case in any hyper-automation project. Both monetary and nonmonetary factors can cause it. Any task must be well-defined, authorized, and closely monitored for any project to be effective. Most hyper-automation attempts fail because they automate the pain points rather than the benefit points.

- Operational complexity:

Hyperautomation is still in its early stages, and any process you plan to automate must be prioritized for hyper-automation depending on operational complexity. For example, a vertical would develop confidence by automating more straightforward procedures initially, then moving on to more complicated ones afterward.

- Process scalability:

The benefits of hyper-automation are also dependent on the process's magnitude. Therefore, Hyperautomation is not feasible for any process that is not scalable.

- Investment in machine learning and AI:

RPA-enabled businesses are more dynamic. Some procedures are performed entirely by robotic tools for these businesses.

Artificial intelligence (AI) and machine learning (ML) aren't leveraged enough at this level, and not everything is automated. Even if 90% of your operations are automated, as many insurance businesses do today, it's probable that humans still handle the remaining 10%. Perhaps the human element is still necessary in this case for claims disputes or sophisticated modifications.

- Cost of implementation failure:

Emerging technology can occasionally give superior results in areas where human judgments rely on complicated parameters. However, before using hyper-automation in critical assessments such as medical advice, you must carefully analyze the costs and dangers of failure and find ways to diminish them.

Organizations have only had mediocre success deploying automation in the recent decade. Automation did not wholly succeed for various reasons, including that the technology was still relatively new and businesses were not ready to take full benefit.

- Choosing the proper use case/process:

Business models in most industries are continually shifting based on their maturity in adopting innovative technology. For example, we have watched technology businesses transform into banks and telecom service providers transform into technology companies as they expanded their offerings to include mobile, data, and cloud services.

As a result, while most verticals would benefit from hyper-automation, it would be advisable to examine the process level that identifies a specific vertical that is ready to aid.

4. When and how can hyper-automation prove to be a game-changer? (Use Case Scenarios of 2022)

The advancement in tech is a never-ending process. It would help if you properly evaluated the input required to invest in technology that helps to boost your organizational goals in the long run.

We are talking about companies that may benefit from hyper-automation in their transformation initiatives if they follow three crucial steps.

The 1st is vision:

The capability and scope to answer the following question, "what will it take to accomplish your goals"?

It will not only help a business to see ahead and forecast. But, to dominate the market first, you have to convince your stakeholders about implementing hyper-automation and its benefits.

Using hyper-automation, you are not adding new elements but boosting the existing process. As a result, managers will have to drop their reliance on legacy applications and learn how the technology works to implement it.

The 2nd step is a clear definition:

Once you set the vision, you must describe the capabilities required to achieve the new image and create a flexible, readily updated, and customer-focused model.

Hyperautomation and advanced technologies, such as RPA, iBPMS, AI, NLP, and decision management, will be used in the model's business operations to supply these operational capabilities and those needed to outpace the competition and acquire market share.

Clearly define all tasks in each business function in the model and automation support capabilities for each activity inside the process.

The process will involve considering critical components such as data, emails, or forms transacted between activities as processes, the amounts of information passed, possible difficulties, performance assessment measures, and strict standards. The model must have sufficient detail for the system to comprehend all operational capabilities.

The 3rd step is execution:

The third step involves executing the vision into action by building and implementing a new state of company operations that will continue to evolve.

This phase determines, defines, and executes the action plan that will take the firm from where it is today to future-standard activities. This is the tricky part. It will require a totally new execution strategy that combines business transformation with hyper-automated application development. While objectives continue to change, most businesses will need help moving forward quickly.

In hyper transformation, the end goal may alter to reflect new technological capabilities, regulatory needs, innovations, and consumer engagement requirements. As a result, firms that tackle transformation in separate business function steps and then mix them will get the actual advantage.

4. The transformative role of hyper-automation across the industry

Let's further learn how hyper-automation performs dynamically across distinct sectors.

Industry	**Role of Hyperautomation**
Customer Support	Using conversational AI and RPA in businesses powered by machine learning to study and understand customer behavior. The data obtained is then analyzed to enhance responses to the most often asked questions. BPOs are already using RPA to automate monotonous time-consuming tasks. However, they need to take it further and hyper-automate using AI and machine learning. This might lead to a significant decrease in customer attrition and improved customer satisfaction.
Financial Transactions	Finance companies face the wrath of strenuous processes that have the potential to streamline easily. It faces numerous frauds and scams but tries its best to keep the money safe. Significantly, the regular step involving the approval of financial transactions can be automated. This will lead to improved efficiency in financial organizations and better management of threats to loss of information.
Claim Processing	One industry that has just started picking up the pace of digital transformation leveraging automation is the insurance industry.

	Claims processing is a headache for customers. However, after automating significant operations claim process handling has improved its image. These operations are automated to boost efficiency in fetching adequate information that helps make quicker decisions.
Document Processing	The majority of the Use Cases across the industry will require documents, emails, contracts, transcripts, etc. (simple and complex) that are unstructured / semi-structured. For the implementation of such use cases, the system needs an end-to-end process that begins with extraction (OCR) > Computer Vision > NLP > Data Keywords > AI/ML to reduce human involvement and make this digital smart. Hyperautomation offers the cognitive ability to perform intelligent document processing. Such an end-to-end automated process empowers the managers to store, fetch, categorize & do more tasks indefatigably.
Lead Generation	Sales are the first door for any Industry & it is becoming more conversational and data-driven. Therefore, the probability of RPA and AI/ML becomes very important to help align the initial workflow to create the product/market awareness and later convert this into a potential engagement. RPA and the sales team are helping them establish the initial awareness through many channels such as Linked-In, Zoho, and Email Campaigns to construct the pipeline and let the salespeople be more focused on the active client engagement. Hence, Hyperautomation is enhancing customer engagement and simultaneously helps salespeople to focus on critical aspects.
Claim Handling	Fluctuating fuel prices, concerns about diesel automobiles' environmental impact, and a continual shift to electric vehicles will increase the frequent EV overheating fire accidents. As a result, the automobile industry will have frequent claim handling cases in the future. Hyper-automation can feasibly help automakers manage this critical phase by optimizing the Claim Handling process. In addition, AI-powered hyper-automation technologies might automate other

	activities are assembly, quality assurance, vehicle maintenance, and quality control. **Consider Tesla and self-driving cars:** AI and cognitive computing, along with improved sensors, have revolutionized the way we think about transportation.

5. How to plan and manage Hyperautomation initiatives?

Once you decide to get on the journey to adopt hyper-automation for your business operations, it is time to learn how to plan and manage hyper-automation initiatives. First, consider the points mentioned below:

1. Define the business objective:

Ask questions. Why do you need it? How will it help you enhance your resolution process? What impact does it have on your goals?

Now, based on the answers, decide the objective. It could be possible that the aim was decided prior, leading to the adoption of advanced technology.

2. Business process analytics:

Utilize the insights from analytics to understand the aspects crucial to automation, such as the time taken to complete a task, type of task, task complexity, availability of data, the scope of leveraging tech, and more.

3. Technology Exploration:

Hyperautomation isn't a solution that works for all. Instead, it combines technologies that have evolved with time and are still improving with mass adoption across industries. So, there is a possibility that your business needs are met just with AI technology and a whole automation transformation won't be needed; choose wisely.

4. Deliver/production:

Defining objectives and achieving won't always align. One has to be as dynamic as the market. Early adopters have seen significant growth in their companies' operating abilities. However, it has worked for those who were constantly enhancing their process according to the elements that fluctuate the market needs.

5. Measure, iterate, reiterate:

To deliver as promised isn't always the right to do. Some uncertain events and aspects significantly impact the market trends. So you have to measure & iterate regularly and sometimes reiterate to keep up with the industry.

6. Govern and scale:

Once you are confident about working with hyper-automation, it is time to scale above and par the market trends and be the catalyst. It can be a strategy or movement that you may want to try and share with the world. So, get on with it, get your hands dirty!

6. Hyperautomation projects are overpromised and underdelivered

According to Gartner, by 2024, over 70% of large global enterprises will have over 70 concurrent hyper-automation initiatives.

The magic of hyper-automation won't work for your business if your foundation is not established. A company lacking proper vision and understanding of its operations mustn't expect machines to streamline it for them. Instead, you have to tell and formulate the process for the device to perform.

It is known as the Resilience Problem. IT resilience is the foundation that paves the way to achieve business goals. In terms of Hyper-automation, REAP(Resilience, Efficiency, Agility, Productivity) criteria are to be fulfilled.

So, how could you make sure that your foundation, aka REAP criteria, is ready to implement hyper-automation? Please consider the points mentioned below to make sure that you deliver the much you promise:

1. **Siloed Operations aren't helpful:** It is crucial to understand that AI is a part of your digital strategy and must be treated like one. When blended with your digital strategy, AI helps in reaching broader strategic goals. So, for AI, you must identify the processes that are backed by data for seamless functioning.
2. **A clear definition is a must:** We have covered this point above. Instead of making the AI use machine learning to figure out on its own, you must have a blueprint about the processes you wish to improve with hyper-automation. The precise definition will make it easy for machine learning's optimal execution.

3. **Integration:** Human intervention is essential for mapping any system be it a CRM, chatbot, or hyper-automation. So, there must be a team to supervise the functioning of hyper-automation that aligns business operations with achieving the best of their potential. To deliver effectively, you must reiterate and reallocate resources by understanding the speed of AI and the prospect of hyper-automation.

4. **Analyze the insights:** Reiteration and reallocation are important as hyper-automation is a continuous process that needs constant monitoring to find the scope of improvement in business operations. Leverage insights gathered by AI to keep up the flow of growth and constant improvement in the hyper-automation cycle.